

A well-formedness condition on alternative generation

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1. Introduction

Listeners often form inferences by negating *alternatives* to an utterance—what might have been said but was not. For instance, from (1a) one infers that John did not do all of the homework, because (1a) has a logically stronger alternative (1b), which the speaker would have asserted had they believed it. Similar reasoning may be used to explain the oddness of (2a), which has (2b) as an alternative. The determiner *both* in (2b) introduces the presupposition that Sue has exactly two arms. From the utterance of (2a) one then infers that the speaker is not committed to this presupposition, contradicting world knowledge.

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|-----|-----|--------------------------------|-----|-----|------------------------------|
| (1) | (a) | John did some of the homework. | (2) | (a) | # Sue broke all of her arms. |
| | (b) | John did all of the homework. | | (b) | Sue broke both of her arms. |

But the nature of alternatives has long been under debate. Which alternatives are generated? And to what extent is their derivation sensitive to syntax? In this paper we provide novel data from which we argue that syntax plays a dual role in constraining alternatives: not only does structural *complexity* constrain alternatives, as argued by Katzir (2007), but syntactic *well-formedness* filters out potential alternatives, so that an inference that could only arise by negating a syntactically ill-formed alternative does not arise.

We base our argument on the distribution of *temporariness inferences* in English. As Dowty (1979) notes, the use of the progressive aspect with a stative verb in (3a) gives rise to an inference, formed by negating a non-progressive alternative, which contradicts world knowledge and yields infelicity. But we show that such inferences do not arise in environments where syntax independently requires that an active verb be progressive, like the reduced relative clause in (3b), even if a more complex full relative clause alternative with a non-progressive verb could in principle be generated, should one reject the structural theory of Katzir (2007). Thus, alternatives must be syntactically constrained as discussed above.

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|-----|-----|--|
| (3) | (a) | # New Orleans is <i>lying</i> at the mouth of the Mississippi River. |
| | | ↘ It is not regularly the case that New Orleans <i>lies</i> at the mouth of the Mississippi River. |
| | (b) | The city <i>lying</i> at the mouth of the Mississippi River is New Orleans. |
| | | ↗ It is not regularly the case that New Orleans <i>lies</i> at the mouth of the Mississippi River. |

Our argument is in tension with the pragmatic literature in two respects. First, it has been argued that syntax does not fully constrain alternative generation—that is, alternatives corresponding to ungrammatical or otherwise unavailable structures *do* play a role in pragmatic competition (Chemla 2007, Buccola et al. 2022). One well-known instance is due to Chemla (2007), who observes that while a language like French lacks a lexical element corresponding to English ‘*both*,’ the oddness of (2a) persists even in French, suggesting that the French version of (2a) must have an alternative that is not expressible in the language (see Jeretič et al. 2024a,b for a discussion). It has also been suggested that the Katzirian notion of structural complexity is not necessary, as the observations that originally motivated it can be derived without it (cf. Schwarz & Wagner 2024, Hirsch 2024).

In this short paper we will not directly engage with this literature. Rather, we focus solely on illustrating that in a particular set of sentences, syntax and structural complexity do play important constraining

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roles, leaving for future work the larger task of distinguishing cases in which they do from those in which they apparently do not.

We proceed as follows: Section 2 presents Dowty’s temporariness inferences. Section 3, introduces the syntactic well-formedness filter and shows how it derives the attested distribution of such inferences. Section 4 rejects an alternative account of that distribution that does not appeal to a well-formedness condition, and Section 5 shows that the syntactic approach makes good predictions. Section 6 concludes.

2. Temporariness inferences

I(ndividual)-level statives are infelicitous in the progressive, which is marked in English by an *-ing* verbal suffix, as shown in (4). S(tage)-level statives, on the other hand, are only *sometimes* infelicitous in the progressive. For example, when one considers the distribution of so-called “positional” verbs like *sit* or *lie*, one finds that the felicity of the progressive depends on the nature of the verb’s argument (5-6). More precisely, when the verb’s argument denotes an entity that is assumed to have a more or less fixed location, the use of a positional verb in the progressive is infelicitous.

- (4) John knows/#is knowing French.
- (5) *Arguments compatible with s-level statives*
 - (a) The guitar lies/is lying under the bed.
 - (b) One corner of the piano rests/is resting on the bottom step.
- (6) *Arguments incompatible with s-level statives*
 - (a) New Orleans lies/#is lying at the mouth of the Mississippi River.
 - (b) That argument rests/#is resting on an invalid assumption.

Because not all statives disallow PROG, Dowty (1979) suggests that we should not encode the incompatibility between the two into the semantics of PROG or of stativity. Rather, he attributes it to a competition between the PROGRESSIVE form of the verb, and its (morphologically unmarked) non-progressive form, henceforth referred to as IMPERFECTIVE (following Deo 2009). Assume a semantics according to which the progressive version of (6a) conveys that New Orleans (NOLA) lies at the mouth of the Mississippi at utterance time (7), whereas its imperfective version conveys that NOLA lies there *regularly* (8). Then, as long as (8) is taken to be an alternative to (7), uttering the PROG version triggers an inference that the IMP one is false; i.e., that NOLA does not *regularly* lie at the mouth of the Mississippi. But, obviously, since cities are unable to move, this inference—which we call, hereafter, the *temporariness inference* (TI) of the progressive—contradicts our world knowledge, giving rise to infelicity.¹

- (7) $\llbracket \text{NOLA is **lying** at the mouth of the Mississippi} \rrbracket^t = 1$ iff NOLA is at the mouth of the Miss. at t .
- (8) $\llbracket \text{NOLA **lies** at the mouth of the Mississippi} \rrbracket^t = 1$ iff it holds at t that NOLA is regularly at the mouth of the Mississippi.

In contrast to cities, guitars *are* movable, so that no infelicity results when PROG is uttered in (5a). Furthermore, when the common ground is such that an otherwise “immovable” subject *is* movable, the infelicity of PROG with a stative verb disappears. This is expected on Dowty’s analysis, since, in such a case, the negation of an IMP sentence with a positional verb *is* compatible with world knowledge. For instance, if the context makes it so that cities are no longer considered immobile, a noun phrase denoting a city may be the subject of the PROG VP in (9).

¹ Our toy treatment of the imperfective predicts that negation as in (1) should have a reading in which John isn’t smoking at some sub-interval of the utterance time interval, assuming that negation scopes above IMP, paraphrased in (2a). But (1) seems to have the strengthened reading in (2b). Adapting von Stechow (1997)’s treatment of all-or-none readings in generic bare plurals, we suggest that strengthening is due to a homogeneity presupposition on IMP. That is, IMP is only defined when either there regularly are p events or if there regularly are not p events.

- (1) John doesn’t smoke.
- (2) (a) It is not regularly the case that John smokes.
- (b) It is regularly the case that John doesn’t smoke. (Strengthening)

- (9) *Due to a catastrophic earthquake*, NOLA is lying on an island in the Gulf of Mexico.

Note, finally, that deriving the TI from pragmatic competition, rather than from the lexical meaning of PROG, is supported by the fact that TIs disappear in discourse contexts when IMP is not relevant; namely, contexts in which the discourse participants are not concerned as to whether a state regularly holds, as in certain “vivid” narratives (10).

- (10) Our ship entered the Gulf, and, behold! NOLA was lying at the mouth of the Mississippi in all its grandeur.
↗ It is not regularly the case that NOLA lies at the mouth of the Mississippi.

On the basis of the properties of TIs introduced above, we adopt Dowty’s approach that derives them as arising from a competition between alternatives.² Next, we show that when syntax rules out the possibility of forming an IMP alternative to a PROG utterance, the expected TIs go away—resulting in the felicity of PROG in environments in which it is otherwise infelicitous—suggesting that syntax “filters out” ill-formed potential alternatives.

3. Missing temporariness inferences

We have seen that when the subject of an s-level stative positional verb is contextually understood to have a fixed location, the use of PROG produces a TI that contradicts world knowledge, resulting in infelicity. Next, we introduce a novel puzzle: the use of PROG does not produce a TI and is felicitous in the three syntactic environments illustrated in (11): reduced relative clauses (RRs), existential *there*-sentences (ES), and VP predicate inversion (PI).³

- (11) *Syntactic environments in which temporariness inferences are missing*
a. NOLA is a city *lying* at the mouth of the Mississippi River. (RR)
b. There is a city *lying* at the mouth of the Mississippi River. (ES)
c. *Lying* at the mouth of the Mississippi River is the city of NOLA. (PI)

We argue that the relevant property unifying these syntactic environments is that they disallow IMP in the active voice (in standard American English), and we propose that TIs do not arise here because alternatives are subject to the *syntactic well-formedness filter* in (12).⁴

- (12) *Syntactic well-formedness filter on pragmatic alternatives*
An alternative must be associated with a convergent syntactic derivation.

We show next how (12) prevents TIs from being generated in (11). Then in Section 4 we defend this approach against an alternative account, on which the *-ing* forms in (11) are not instances of PROG, and thus not predicted to give rise to TIs to begin with.

² We are not concerned here with how exactly competition gives rise to the TI of PROG; upon certain assumptions, it could be implemented as an implicature, or perhaps as an anti-presupposition effect. What is important here is that under any satisfactory account we must assume that a sentence with PROG has as its alternative the corresponding sentence with IMP. Note, further, that this assumption gives rise to a potential “symmetry problem:” what prevents PROG from being negated when IMP is used? We hint at two possible approaches. First, we assume here that the obligatory auxiliary *be* used with PROG makes PROG sentences strictly more complex than their IMP counterparts, so that the well-formedness condition (12) and the complexity condition on alternative generation (14), both introduced below, together prevent PROG from being an alternative to IMP. Another approach is to adopt lexical entries such that IMP asymmetrically entails PROG, perhaps along the lines of Deo (2009).

³ These cases do not require a special discourse context of the form discussed in connection with (9) or (10).

⁴ The formulation in (12) is intentionally agnostic about the ontology of alternatives. If alternatives are viewed as essentially syntactic, then (12) says that such “structural” alternatives must be syntactically well-formed. If, on the other hand, alternatives are viewed as logical objects (e.g., propositions), then (12) states that a potential alternative is accessible to pragmatic reasoning only if it is the interpretation of a syntactically well-formed structure.

3.1. Relative clauses: Full vs. reduced

Support for the existence of a syntactic well-formedness filter comes from the interaction of verb form (PROG vs. IMP) and clause size (full vs. reduced) in relativization. As in main clauses, the use of PROG for s-level statives in full relative clauses gives rise to a TI (13ai), and is therefore infelicitous, whereas the use of IMP is fully acceptable (13aai). In contrast, in reduced relative clauses, which are characterized by the absence of tense or complementizer material, PROG may be used for s-level statives without giving rise to a TI (13bi). In fact, this is the only possibility for active voice verbs in reduced relatives: IMP forms are simply ungrammatical (13bii).⁵

- (13) (a) *Full relative clauses*
- (i) # NOLA is a city that/which is lying at the mouth of the Mississippi River.
 - (ii) NOLA is a city that/which lies at the mouth of the Mississippi River.
- (b) *Reduced relative clauses*
- (i) NOLA is a city lying at the mouth of the Mississippi River.
 - (ii) * NOLA is a city *lie(-*s)* at the mouth of the Mississippi River.

In explaining the pattern exhibited in (13), we present an analysis in which IMP is a valid alternative to PROG in full relative clauses, but not in reduced relative clauses. We begin by adopting the proposal of Katzir (2007) and Fox & Katzir (2011) in analyzing alternative propositions as always being associated with alternative syntactic *structures* derivable via substitution and structural simplification from the uttered sentence; a version of this condition sufficient for our purposes is presented in (14).

- (14) *Complexity condition on structural alternatives* (Katzir 2007, Fox & Katzir 2011, et seq.)
A sentence Φ may have Φ' as a structural alternative only if Φ' may be derived from Φ by iterated operations of replacing a tree by a sub-tree or by substituting one lexical item for another.

Given the two conditions (12) and (14), let us first see how IMP is an alternative to PROG in full relative clauses like those in (13a). We assume that the structure of a full relative clause with PROG is as in (15i), ignoring head movement (e.g., of [_{AUX} be] to T) and irrelevant structural details. Replacement of the tree headed by Aux with the subtree of it that is headed by Asp yields (ii), and subsequent substitution of [_{ASP} IMP $\emptyset$] for [_{ASP} PROG-ing] yields (iii), namely, the structure of a full relative clause with an IMP main verb. Because (iii) is derived by only by subtree replacement and substitution of lexical items, (14) is satisfied; and because (iii) is grammatical, the well-formedness filter (12) is satisfied. Then the interpretation of (iii) yields an IMP alternative to the PROG preadjacent, the negation of which yields a TI.

- (15) *Derivation of IMP from PROG in a full (tensed) clause*
- (i) [_{CP} which PRES [_{AUXP} [_{AUX} be] [_{ASPP} [_{ASP} PROG-ing] [_{VP} lie at the mouth of the Mississippi]]]]
 - (ii) [_{CP} which PRES [_{ASPP} [_{ASP} PROG-ing] [_{VP} lie at the mouth of the Mississippi]]]
 - (iii) [_{CP} which PRES [_{ASPP} [_{ASP} IMP $\emptyset$] [_{VP} lie at the mouth of the Mississippi]]]

To see the relevance of the well-formedness condition (12), we must consider a case where the complexity condition is met, but well-formedness is not. To divorce the two to begin with, of course, we must assume that the Katzir algorithm can generate syntactically ill-formed alternatives that can nevertheless be interpreted. While one might think that the algorithm has a syntactic ill-formedness condition already built into it, as far as we can tell this has not been made explicit in the literature; in principle, we can view the output of the algorithm as a set of denotations, derived from the manipulation of structures that are not subject to considerations of the “narrow syntax”. If that is the case, then the two conditions come apart in the cases at hand. The novel contribution here is to explicitly consider the role of narrow syntax in alternative generation.

Take reduced relatives (RRs), for instance. Following Harwood (2018), we take them to involve less structure than full relatives. For concreteness, assume that a RR is simply an AspP. From the well-formed PROG relative clause in (16i), replacement of [_{ASP} PROG-ing] by [_{ASP} IMP $\emptyset$] yields the ungrammatical (ii).

⁵ In some varieties of English, the full/reduced dichotomy is less clear, and tensed forms may be used in sentences like *NOLA is a city lies at the mouth of the Mississippi River*. We do not address such constructions here.

(16) *Derivation of ungrammatical IMP from PROG in a reduced clause*

- (i) [_{ASPP} OP [_{ASP} PROG-*ing*] [_{VP} lie at the mouth of the Mississippi]]
- (ii) *[[_{ASPP} OP [_{ASP} IMP_Ø] [_{VP} lie at the mouth of the Mississippi]]

Though the precise reason for the ungrammaticality of (16ii) is beyond the scope of this short paper, we assume that it is syntactic in nature, rather than semantic.⁶ Consider, then, the applicability of the two conditions on alternatives that we have discussed. The complexity condition prevents an IMP full relative clause like (13aii) from being a structural alternative to the PROG reduced relative in (13bi), since deriving a full relative clause from a reduced one would require insertion, which is not one of the operations permitted by (14). But it is the additional condition of syntactic well-formedness which prevents the ill-formed representation in (16ii) from being a licit structural alternative of (16i).

3.2. *Existential there sentences*

A second construction in which the use of PROG does not give rise to the expected TI is that of verbal existential sentences, which are characterized an expletive *there* subject and an immediately preverbal noun phrase pivot, exemplified in (17a).⁷ Existential *there* sentences differ from other sentences with an existential interpretation in that the use of PROG in the latter gives rise to a TI (18).

(17) *Existential sentences with an expletive there subject*

- a. There is a well-known city *lying* at the mouth of the Mississippi River.
- b. * There is a well-known city *lie(-*s)* at the mouth of the Mississippi River.

(18) *Existential sentences with a noun phrase subject*

A well-known city *lies/#is lying* at the mouth of the Mississippi River.

Like reduced relative clauses, but unlike main clauses, the form of the verb in *there* existentials must be participial (i.e., a form that can be the complement of auxiliary *be*), lest ungrammaticality result (17b). Thus, the participial *-ing* verb forms that realize PROG are permitted, but not the zero-derived verb forms that occur in IMP. The same reasoning presented in the previous section for reduced relative clauses is then applicable here in the case of *there* sentences. That is, the well-formedness condition prevents the IMP sentence in (17b) from being the basis for an alternative whose negation would yield a TI.

One also finds expletive *there* as the subject of *presentational* constructions, in which the DP pivot occurs in a postverbal position and is focused (Aissen 1975). Both IMP and PROG forms are grammatical here; the use of PROG is felicitous provided that temporariness does not contradict world knowledge, as in (19), where it is understood that medical instruments are liable to move.

(19) There are lying on the table various kinds of medical instruments.

In (20) we see that IMP is grammatical in presentational sentences. Then what our account predicts is that, because IMP is a valid alternative to PROG (by virtue of satisfying conditions (12) and (14)), a TI should appear when PROG is used. This is in fact what we find.

(20) There *lies/#is lying* at the mouth of the Mississippi River a famous city.

⁶ The ban on IMP forms in reduced relatives is related to a cross-linguistic property of reduced relatives that their verb must be nominal, as reflected by gender inflection, for instance (Iatridou et al. 2001, Thoms & Walkden 2019).

⁷ We follow Milsark (1974) in analyzing verbal existential sentences as not involving reduced relative clauses. Notice in particular that whereas relative clauses prohibit extraction (I), existential constructions permit it (II), indicating that the lexical verb in an existential sentence is the main verb, not the head of an adjunct to the pivot noun phrase.

- (I) (a) I read a pamphlet lying on the table. (b) *What did you read a pamphlet lying on?
- (II) (a) There was a pamphlet lying on the table. (b) What was there a pamphlet lying on?

In (III) we see that movement from an s-level stative PROG verb phrase is acceptable, indicating that the absence of TIs in such constructions is not somehow due to a relative clause analysis. We therefore take verbal existentials and reduced relative clauses to be distinct constructions in which TIs are absent.

(III) [What river]₁ is there not_{FOC} a city lying at the mouth of —₁? ↗ TI

3.3. VP predicate inversion

In VP predicate inversion constructions, the apparent subject of the sentence is a participial verb phrase and the noun phrase “would-be” subject occurs in a postverbal focus position, much as in presentational *there* sentences (Emonds 1970, Samko 2016). Inversion is typically used to introduce a new discourse referent, so that (21a) be used felicitously, for instance, as the first sentence of an article about NOLA, as may its IMP uninverted counterpart (21b). TIs are missing in predicate inversion.

- (21) *Missing temporariness inference in VP predicate inversion*
- a. Lying at the mouth of the Mississippi River is the city of NOLA. (PI)
 - b. The city of NOLA lies/#is lying at the mouth of the Mississippi River.

Notice that structural simplifications and lexical replacements on the syntactic representation of progressive (21a) so as to yield an IMP structural alternative satisfying the complexity condition (14) could only result in various ungrammatical outcomes (22). The well-formedness filter then prevents an LF corresponding to a version of (22) from being a valid structural alternative.

- (22) *Lie(-*s) at the mouth of the Mississippi River is/*does the city of NOLA.*

The use of IMP in an uninverted sentence like (21b) is of course grammatical, but it may be shown that condition (14) prevents it from being a valid structural alternative to an inverted sentence like (21a). To do so, assume, following Samko (2016), that inversion constructions involve the VP in specTP and the DP in a lower specifier position; to derive (21b) from (21a), one would need to swap VP and NP, which is not possible by structural simplification and lexical replacement alone.

We have seen three cases where the use of an *-ing* form fails to produce a TI, unlike the typical cases discussed in Section 2. Our analysis has depended throughout on the assumption that the *-ing* forms in question are genuine instantiations of PROG. We argue next that this assumption is valid.

4. Distinguishing PROG *-ing* from gerundive *-ing*

What Huddleston & Pullum (2002) call the “gerund-participle” marker *-ing* does more than form PROG verbs; we illustrate three of its other functions in (23). *Adjectival* and *nominal gerundive -ing* are diagnosed by their inability to take a DP complement (23a-b). The suffix *-ing* may also derive gerunds of mixed categorial status (23c), which are verb-like in taking DP complements, but noun-like (or determiner-like) in heading a phrase that occurs in argument positions and that saturates predicates.

- (23) *Non-progressive uses of -ing*
- (a) The speech was very inspiring *(to) me. (Adjectival *-ing*)
 - (b) The indiscriminate killing *(of) civilians is criminal. (Nominal gerundive *-ing*)
 - (c) Continuously entertaining *(of) guests is tedious. (Mixed gerundive *-ing*).

The uses of *-ing* illustrated in (23) do not contribute progressive meaning, and so are not expected to give rise to TIs. Our argument for the proposed well-formedness filter (12) thus crucially rests on the as-yet unjustified assumption that the environments when TIs disappear necessarily involve PROG *-ing* rather than some other gerundive *-ing*. Below, we provide three arguments that *-ing* in these environments is genuine PROG, using diagnostics from (i) the unavailability of perfective readings, (ii) the availability of wide scope, (iii) and connectivity effects in morphological selection.

4.1. Unavailability of perfective readings

Progressive aspect is a species of imperfective, presenting an eventuality as being incomplete.⁸ Incompatibility with perfective interpretations is then a diagnostic for an instance of *-ing* being an exponent of PROG rather than a non-progressive gerundive marker. Indeed, we find that gerunds freely

⁸ Throughout this paper, when we contrast PROG with IMP, the latter denotes the non-progressive imperfective.

permit perfective interpretations, as in (24). But perfective readings are unavailable in RRs (25a), PI (25b), and ES (25c), patterning with ordinary progressive VPs.

- (24) Writing that book was a great achievement. (Mixed Gerundive *-ing*)
 ≈ It was a great achievement to write/have written that book.
- (25) *For years I thought that Newton_F wrote that book, but ...*
 Leibniz_F wrote/#was writing that book.
- (a) #The person writing that book was Leibniz_F. (RR)
 (b) #Writing that book was Leibniz_F. (PI)
 (c) #There was [someone else]_F writing that book. (ES)

However, if a suitable reference time is available to support the “in progress” interpretation of the progressive (e.g., supplied by a frame adverbial), RR, PI, and ES become felicitous (26).

- (26) *For years I thought that Newton_F wrote that book, but when I stepped out of the time machine ...*
 Leibniz_F was writing that book.
- (a) The person writing that book was Leibniz_F. (RR)
 (b) Writing that book was Leibniz_F. (PI)
 (c) There was [someone else]_F writing that book. (ES)

The unavailability of perfective readings and the availability of the “in progress” reading in RR, PI, ES together suggest that these constructions involve genuine PROG *-ing*.

4.2. Availability of wide-scope for quantifiers within the *ing*-phrase

We next present an argument that the *-ing* form in VP predicate inversion in particular is the head of the main AspP of the clause. We begin with an observation by Thoms & Walkden (2019): a VP-internal quantificational phrase (QP) (e.g., *every bank*) may take either wide or narrow scope with respect to a VP-external QP (e.g., *a police officer*) both in uninverted sentences (27) and in inverted sentences (28).

- (27) A police officer was guarding every bank. (✓ ∃∀; ✓ ∀∃)
 (28) Guarding every bank was a police officer. (PI; ✓ ∃∀; ✓ ∀∃)

We observe, however, that, a quantifier contained within a *gerundive -ing* subject must take narrow scope with respect to a quantifier outside of the subject (29).

- (29) Guarding every bank exhausted a police officer. (Mixed gerundive *-ing*; ✓ ∃∀; ✗ ∀∃)

To explain the contrast between (28) and (29), assume that wide scope of QP₁ over QP₂ requires an LF configuration in which QP₁ c-commands QP₂. To derive the ∀∃ reading of (27), the predicate-internal QP₁ undergoes QR at LF to a position c-commanding the predicate-external QP₂ (30). The PI sentence (28) has a similar LF (31), with two relevant differences: (i) PI does not involve movement of QP₂ to SpecTP, and (ii) VP moves to specTP, reconstructing at LF (as indicated by the strike-through).

- (30) *Wide scope of predicate-internal quantifiers in uninverted sentences*

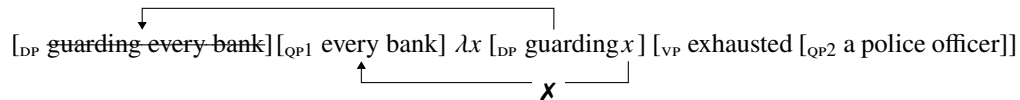
[_{QP1} every bank] λx [_{QP2} a police officer] λy was y guarding x

- (31) *Wide scope of predicate-internal quantifiers in inverted sentences*

[_{VP} ~~guarding every bank~~] [_{QP1} every bank] λx was [_{QP2} a police officer] [_{VP} ~~guarding x~~]

As for gerundive DP subjects like that in (29), we follow Charlow (2010) in analyzing DP as a scope island, so that QR out of gerunds simply cannot occur to create wide scope. Thus the derivation of wide scope for the gerund-internal QP₁ in (32) is ill-formed.

(32) *Illicit QR out of gerund*



The availability of wide scope for predicate-internal quantifiers thus distinguishes PROG *-ing* forms from gerundive ones, and indicates that sentences like (33) necessarily involve a PROG VP.

(33) Lying on every tributary of the Mississippi is a town with its own unique culture. (PI; $\checkmark \forall \exists$)

No TI arises in (33) despite wide scope of the predicate-internal quantifier forcing a PROG analysis of the *-ing*. We therefore conclude that the absence of TIs in PI cannot be due to the fronted constituent being a gerund; rather, it is a progressive VP predicate (AspP).

4.3. Connectivity in morphological selection

Continuing to focus on PI, our final argument for the *-ing* form being a genuine realization of PROG concerns connectivity effects in morphological selection. Observe first that the only verbal heads that select for a complement headed by PROG are forms of *be*.⁹ In (34b) we see, for instance, that the AspP *entertaining the guests* may not be the complement of the lexical verb *seem*.

(34) *PROG -ing must not head a complement of a lexical verb*

- (a) John seems to *be* entertaining the guests. (b) *John *seems* entertaining the guests.

A gerundive *-ing* phrase, on the other hand, *may* be the complement of a lexical verb, for instance, *remember* in (35I). Moreover, in (II) we see that gerundive *-ing* may project to a DP argument of a predicate, here *tedious*, with or without a form of *be*.

(35) *Gerundive -ing may head a complement of a lexical verb*

- (I) John remembers entertaining the guests.
(II) (a) Entertaining the guests seems to be tedious.
(b) Entertaining the guests seems tedious.

In contrast to gerundive *-ing*, the phrase projected by *-ing* in PI *must* co-occur with a form *be* (36), as originally observed by Emonds (1970).

(36) *The -ing form of predicate inversion must cooccur with be*

Mark is cooking, Claudia is getting drinks, and ...

- (a) ...entertaining the guests seems to be John_F.
(b) * ...entertaining the guests seems John_F.

The ungrammaticality of PI in (36b) is explained if it has the same underlying structure as its uninverted counterpart (34b). In the pre-movement configuration (37), AspP headed by PROG is illicitly the complement of the lexical verb *seem*; subsequent movement of AspP to specTP to yield an inverted sentence does not alleviate this situation.

(37) *[_{VP} John [_{V'} seem [_{AspP} -ing [_{VP} entertain the guests]]]]

This concludes our discussion of diagnostics for PROG versus gerundive *-ing* forms. The first, concerning the unavailability of perfective readings, diagnosed predicate inversion, reduced relatives, and existential sentences as involving genuine progressive aspect. Our second and third arguments, from scopal possibilities and connectivity in morphological selection, respectively, corroborated this assertion for predicate inversion constructions in particular. We conclude that the absence of TIs in these three constructions cannot be due to the absence of PROG. Next, we consider extensions of the analysis.

⁹ That is, the English lexicon is such that the configuration [_{X'} X [_{V, *PROG*}] YP [_{PROG}]] is well-formed only if X is *be*, where flanking dots indicate the feature that is selected-for. This formulation does not prevent PROG from occurring in configurations where it is not the complement of a verbal head. One such configuration occurs in reduced relative clauses, which we have argued above are maximal projections of PROG and occur as adjuncts to a noun.

5. Extensions

We have argued that the syntactic well-formedness filter (12) prevents TIs from arising in reduced relative clauses, predicate inversion, and existential sentences, which share in common the property that, in the active voice, non-progressive IMP forms are ungrammatical. In this brief section, we extend the analysis to i-level predicates and passive progressives of statives.

5.1. I-level predicates

In this paper we have concentrated our attention on the distribution of s-level statives. But let us return to the observation that i-level statives are infelicitous in the progressive (4). Our observation that TIs are not generated in the syntactic environments that disallow IMP extends to these cases as well—as long as we control for two confounds.

First, predicate inversion is generally only permissible with i-level predicates in the presence of additive focus particles like *also* or *even*, or otherwise comparative morphology, as illustrated in (38).

- (38) I watched some movies today. . .
(a) *Good was ‘Barbie.’
(b) ✓ ‘Barbie’ was good. Also good/Better was ‘Oppenheimer.’

Second, i-level predicates are bad in *there*-constructions in general (39), the discussion of which is beyond the scope of this paper (Milsark 1974).

- (39) *There are men tall.

Controlling for the above, otherwise infelicitous uses of the progressive with i-level statives become felicitous in reduced relatives and predicate inversion (40).

- (40) (a) A student *knowing* French would get by well in this school. (RR)
(b) We all have decent French, but *knowing* French better than any of us is John. (PI)
(c) *There is a student *knowing* French. (ES; explained independently by Milsark’s restriction)

Crucially, neither (40a) nor (40b) require a special context in which the state of x knowing French is temporary. As with s-level statives, the use of PROG with i-level statives seems to be felicitous when the corresponding IMP construction is ill-formed, precisely what is predicted by our analysis.

5.2. Passive progressive forms of stative predicates

Our analysis makes a further prediction: if the constraint banning IMP may somehow be obviated in the syntactic environments under discussion, the use of PROG should once again imply the negation of IMP, yielding a TI and—in the case of statives—resulting in infelicity. In our discussion of presentational *there*-sentences, we have already seen one such case, where the grammaticality of IMP makes the use of PROG infelicitous. We suggest that this also occurs when passive voice is substituted for active voice in reduced relatives and predicate inversion. In (41) we see that (a) in reduced relatives, (b) in predicate inversion, (c) and in existential sentences, the use of IMP with s-level statives is grammatical and felicitous.¹⁰ The use of PROG in these constructions is only felicitous if there is an eventive interpretation, which we assume to involve different lexical items and/or structure from statives.

- (41) (a) The house seated/#*being seated* at the top of the hill is John’s. (RR)
(b) Seated/#*being seated* at the top of the hill is John’s house. (PI)
(c) There is a house seated/#*being seated* at the top of the hill. (ES)

In summary, we find that TIs reappear in reduced relatives, predicate inversion constructions, and *there*-sentences precisely when there is a grammatical IMP alternative to PROG.

¹⁰We use s-level statives as the use of passive voice is somewhat awkward with i-level ones. The point may still be made, however, by consideration of minimal pairs like *known/#being known best are English and French*.

6. Conclusion

In this paper, we have presented a novel pattern of syntactically-conditioned inferences that is best explained by building upon the theory of structural alternatives by introducing a condition according to which potential alternatives are subject to a syntactic well-formedness filter. As mentioned in the introduction, our findings complicate the empirical and the theoretical picture. Future work should attempt to reconcile these various cases into a coherent theory of alternatives.

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